

ECO 146: INTERMEDIATE MICROECONOMIC ANALYSIS

Summer 2 2026, Lehigh University

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Monday -- Thursday, 12pm – 1:35pm

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Office hours: Monday -- Thursday 1:35 pm - 2:20pm, or by appointment

Course Description: This course provides an applied analysis of intermediate microeconomics. The course begins with the tools economists use to model individual and firm behavior, including marginal analysis, optimization, supply and demand, and comparative statics. We then study consumer choice, individual and market demand, production, costs, competitive markets, market power, pricing strategies, oligopoly, and introductory game theory. A central goal of the course is to help students use economic models to analyze real-world markets, policy changes, and firm decisions.

Course Objectives:

At the completion of this course, students will be able to:

1. Apply marginal analysis to consumer and firm decision making.
2. Use constrained optimization to analyze choices made by consumers and firms.
3. Use supply and demand models to analyze competitive market outcomes.
4. Derive and interpret individual demand, market demand, and consumer surplus.
5. Explain the relationships between production, costs, and firm supply decisions.
6. Analyze market outcomes under perfect competition, monopoly, and oligopoly.
7. Use basic game-theoretic concepts to study strategic interactions among firms

Required Text: Robert Pindyck and Daniel Rubinfeld, Microeconomics, 9th edition.

ISBN-13: 9780136879572

(The textbook has been submitted to the Lehigh University Bookstore as a required course material for ECO 146-011.

Students should check the Lehigh Bookstore/course materials portal for available purchase, rental, or eText options.)

Prerequisites: ECO 001 and one of the following mathematics prerequisites: MATH 021, MATH 031, MATH 051, MATH 062, MATH 096, MATH 081, or both MATH 075 and MATH 076.

Students are expected to be comfortable with basic algebra, graphs, derivatives, and marginal analysis. A brief math and calculus review will be provided at the beginning of the course.

Letter Grade Assignment. Tentative, will lower letter cut offs if class average is lower than 80%

100 – 94	A
94 – 90	A-
90 – 87	B+
87 – 84	B
84 – 80	B-
80 – 77	C+
77 – 74	C
74 – 70	C-
70 – 67	D+
67 – 64	D
64 – 60	D-
Below 60	F

Graded items and weights:

Homework 20%

Quizzes 20%

Attendance 5%

Midterm Exam 25% (Covers Chapters 1–4 and will be held during regular class time on Tuesday, July 21.)

Final Exam 30% (Covers Chapters 6–13 and will be held 12:00pm – 1:35pm on Friday, August 14. Not cumulative.)

Course-Specific Academic Integrity Rules

• **Academic Integrity and Respect:** Lehigh University expects that all students will act in a manner that reflects personal and intellectual honesty.

• Cheating includes but is not limited to:

1. The use of any unauthorized assistance in taking quizzes, tests, or examinations. The possession at any quiz or examination of any articles that are prohibited will be regarded as evidence of responsibility.
2. The dependence upon the aid of sources beyond those authorized by the instructor in writing papers, preparing reports or homework, solving problems, or carrying out other assignments.
3. The acquisition, without permission, of tests or other academic material belonging to a member of the university faculty or staff.
4. Any attempt to falsify an assigned grade in an examination, quiz, report, program, grade book, or any other record or document.
5. The creation and/or submission of falsified data in any experiment, research paper, laboratory assignment, or other assignment.
6. Collusion occurs when students willfully give or receive unauthorized or unacknowledged assistance. Both parties to the collusion are considered responsible.

• Plagiarism includes but is not limited to the direct use or paraphrase, of the work, themes or ideas, of another person without full and clear acknowledgement. For further information please refer to <https://citl.lehigh.edu/academic-integrity-resources>

• For this course, unauthorized assistance includes but is not limited to receiving help from other students, tutors, online forums, messaging platforms, file-sharing sites, or AI tools during quizzes or exams. Quizzes, the midterm exam, and the final exam must be completed independently. Although quizzes and exams are open book/open notes unless otherwise stated, students may use only the textbook, course slides, their own notes, a calculator, and materials explicitly permitted by the instructor.

Homework assignments are intended to help students practice and learn the course material. Students may consult the textbook, course slides, and their own notes, and they may discuss general course concepts with classmates. However, each student must prepare and submit their own work. Copying another student's work, allowing another student to copy your work, submitting jointly written answers, or submitting answers generated by AI tools is not permitted.

• Use of generative AI tools, including but not limited to ChatGPT, Claude, Gemini, Copilot, or similar tools, is not permitted during quizzes or exams. Submitting answers produced by another person or by an AI tool as your own work is a violation of academic integrity.

• For Zoom-based exams, students must join the Zoom meeting on time and keep their cameras on throughout the exam. Failure to follow exam instructions may be treated as an academic integrity concern.

Accommodations for Students with Disabilities: If you have a disability for which you are or may be requesting accommodations, please contact both your instructor and the Office of Academic Support Services, Williams Hall - Suite 30, (610-758-4152) or at <https://studentaffairs.lehigh.edu/disabilities> as early as possible in the semester. You must have documentation from the Academic Support Services office before accommodations can be granted.

Course Requirements and Assessment Policies

- **Homework.** There are five homework sets in this course. The due dates are listed in the course schedule below. No late homework will be accepted. Lowest homework score will be dropped.

Homework assignments may include multiple types of questions. Multiple-choice questions should be completed directly on CourseSite. For calculation questions, students may either type their answers directly on CourseSite or upload clear written work showing their calculations. Questions that require graphs or diagrams must be uploaded as written work.

For uploaded work, students should submit clear and readable photos or scans as a PDF, JPG (JPEG), or PNG file. Homework assignments have no time limit, but all required parts must be completed and submitted by the posted deadline. Students may resubmit before the deadline if they need to correct or replace a submission. Only the latest submission received before the deadline will be graded.

- **Quizzes.** There are four graded quizzes in this course: Quiz 1, Quiz 3, Quiz 4, and Quiz 5. Each quiz will be based on the corresponding homework assignment and material covered in class. There will be no Quiz 2 because Homework 2 is followed directly by the midterm exam. Quiz dates are listed in the course schedule below.

Each quiz has a 60-minute time limit and allows only one attempt. Once a student begins a quiz, the timer will continue to run, and the quiz must be completed in that attempt. Students should make sure they have a stable internet connection and enough uninterrupted time before starting the quiz.

- **Attendance and Participation.** This course is officially scheduled as a synchronous online course, and students are encouraged and expected to attend the live Zoom sessions when possible.

Attendance and participation account for 5% of the course grade. Students may earn participation credit for each class session in one of two ways:

1. by attending the live Zoom session, or
2. by watching the class recording and completing the corresponding Attendance Exercise (AE).

Participation credit will be recorded through the Attendance Exercise (AE) scores on CourseSite. Students who attend a live Zoom session do not need to complete the AE for that class session. They will automatically receive full credit for that day's AE. Students who do not attend the live session should watch the class recording and complete the corresponding AE by the posted deadline.

Attendance Exercises are short, low-stakes activities based on lecture content. Each AE will typically include multiple-choice questions and is designed to help students stay on track with the course material. AEs will be posted on CourseSite after class on the same day and will be due two days later. Unless otherwise noted, AEs have no time limit and may be attempted multiple times before the deadline. Late AEs will not be accepted.

- **Midterm Exam.** The midterm exam will cover Chapters 1–4. The midterm will be held from 12pm to 1:35pm on Tuesday, July 21, via Zoom. Students must join the Zoom meeting on time and keep their cameras on throughout the exam.

There will be no make-ups given for the midterm exam. If you have a University-excused absence that has been verified by the Dean of Students, then the value of your final exam will be increased to 55%.

- **Final Exam.** The final exam will cover Chapters 6–13. The final exam is not cumulative and will be held from 12pm to 1:35pm on Friday, August 14, via Zoom. Students must join the Zoom meeting on time and keep their cameras on throughout the exam.

There will be no make-ups given for the final exam. If you have a verified University-excused absence for the final exam, the only option will be to take an alternative version of the final exam from 12pm to 1:35pm on Saturday, August 15.

- **Exam and Quiz Rules.** You may use a calculator, the textbook, course slides, and your own notes for quizzes and exams. However, all work must be completed independently. Giving or receiving help during quizzes or exams, including help from other people, messaging platforms, online forums, or AI tools, is a violation of academic integrity.

Tentative Course Schedule

The following schedule is tentative and may be adjusted depending on the pace of the course. Any changes to topics, assignments, quizzes, or due dates will be announced in class and/or on CourseSite.

Week	Class & Date		Schedule	Assignments	DUE
1	Monday Class 1	7/06	Intro and math/calculus Review		
	Tuesday Class 2	7/07	Ch. 1 – Preliminaries & Ch. 2, Supply & Demand		
	Wednesday Class 3	7/08	Ch.2 Supply and Demand: elasticity and comparative statics		
	Thursday Class 4	7/09	Ch.2 applications: price controls, taxes, consumer and producer surplus	HW #1 Assigned (covers ch1 and 2) at 12pm on CourseSite (CS)	
	Sunday	7/12			HW #1 due at 12pm on CS
2	Monday Class 5	7/13	Ch. 3 Consumer Behavior: preferences, utility, budget constraints;	Quiz 1 (based on HW #1 and class materials) posted by 12pm on CS	
	Tuesday Class 6	7/14	Ch. 3 Consumer Behavior: marginal rate of substitution and utility maximization		Quiz 1 due at 12pm on CS
	Wednesday Class 7	7/15	Ch. 3 Consumer Behavior: consumer choice applications; price and income changes		
	Thursday Class 8	7/16	Ch. 4 Individual and Market Demand: individual demand, income/substitution effects, market demand	HW #2 Assigned (covers chapters 3 and 4) at 12pm on CS Practice midterm questions posted at 12pm on CS	
	Sunday	7/19		Midterm practice answer key posted at 12pm on CS	HW #2 due at 12pm on CS
3	Monday Review 1	7/20	Midterm review: Ch. 1–4	60-minute Review; 30-minute & office hour Q&A	
	Tuesday Midterm	7/21	Midterm Exam: Ch. 1–4	Midterm held during class time. Keep your camera on throughout the exam.	
	Wednesday Class 9	7/22	Ch. 6 Production: production functions, marginal product, average product	No Quiz 2	
	Thursday Class 10	7/23	Ch. 6 Production continued; Ch. 7 Cost of Production	HM #3 Assigned (covers chapters 6 and 7(part)) at 12pm on CS	
	Friday	7/24			Last day to withdraw with a W
	Sunday	7/26			HW #3 due at 12pm on CS

4	Monday Class 11	7/27	Ch. 7 Cost of Production: short-run and long-run costs;	Quiz 3 (based on HW #3 and class materials) posted by 12pm on CS	
	Tuesday Class 12	7/28	Ch. 8 Profit Maximization and Competitive Supply: profit maximization, $MR = MC$		Quiz 3 due at 12pm on CS
	Wednesday Class 13	7/29	Ch. 8 Competitive Supply: short-run supply curve, long-run competitive equilibrium		
	Thursday Class 14	7/30	Ch. 9 Analysis of Competitive Markets: welfare, taxes, price controls, policy analysis	HW #4 Assigned (covers chapters 7(part) 8 and 9(part)) at 12pm on CS	
	Sunday	8/02			HW #4 due at 12pm on CS
5	Monday Class 15	8/03	Ch. 9 Analysis of Competitive Markets continued	Quiz 4 (based on HW #4 and class materials) posted by 12pm on CS	
	Tuesday Class 16	8/04	Ch. 10 Market Power: Monopoly and Monopsony		Quiz 4 due at 12pm on CS
	Wednesday Class 17	8/05	Ch. 10 Monopoly continued; Ch. 11 Pricing with Market Power		
	Thursday Class 18	8/06	Ch. 11 Pricing with Market Power continued; Ch. 12 Monopolistic Competition and Oligopoly	HW #5 Assigned (covers chapters 9(part) 10 11 and 12 (part)) at 12pm on CS. Final practice questions, Part I posted on CS: Ch. 6–11	
	Sunday	8/09			HW #5 due at 12pm on CS
6	Monday Class 19	8/10	Ch. 12 Monopolistic Competition and Oligopoly continued	Quiz 5 (based on HW #5 and class materials) posted by 12pm on CS	
	Tuesday Class 20	8/11	Ch. 13 Game Theory and Competitive Strategy: dominant strategies, Nash equilibrium	Final practice Part I answer key posted on CS Final practice Part II posted on CS: Ch 12-13	Quiz 5 due by 12pm on CS
	Wednesday Class 21	8/12	Ch. 13 Game Theory continued; Final exam review 1 (if have time)	Final practice Part II answer key posted by 12pm on CS	
	Thursday Review 2	8/13	Final exam review 2	60-minute Review; 30-minute & office hour Q&A	
	Friday	8/14	Final Exam: Ch. 6–13	Final exam held on Zoom 12:00pm – 1:35pm; Camera must remain on throughout the exam.	